A.6 Robot Arm Calligraphy

A planar robot arm with two rigid segments of lengths l_1 and l_2 can move in the plane by rotating two joints with angles θ_1 and θ_2. The position of the end-effector (the tip of the robot arm) is given by

$$x = l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2), \quad y = l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2).$$

Given a target point (x, y), the robot must determine the joint angles (θ_1, θ_2) that place the end-effector at that location. This problem is known as *inverse kinematics*. Since the equations are nonlinear, numerical methods are required to compute the angles.

Mission

1. Implement the forward kinematics equations that map (θ_1, θ_2) to the position (x, y) of the robot arm.
2. Given a target point (x, y), formulate the inverse kinematics problem as a nonlinear system and solve it using Newton's method.
3. Choose several points that trace a curve representing "MAT 264 ROCKS". Use interpolation (for example piecewise linear interpolation or spline interpolation) to generate a smooth sequence of points along the curve.
4. For each point on the curve, compute the corresponding joint angles (θ_1, θ_2) using your numerical solver.
5. Simulate the motion of the robot arm and visualize the trajectory of the end-effector. Verify that the robot arm traces the intended curve.
6. Create a simple animation showing the robot arm moving and writing the curve.